Which extraction protocol should I use for my historical freshwater fish samples?
A performance analysis on widley used methods

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* extraction from muscle tissue (otherwise from gill arches)
afl = average fragment length according to TapeStation

	QO (Qiagen Old)	QF (Qiagen Freezing Step)	GENial (magnetic particles)	Promega Maxwell	Dabney & Rohland	PCI
Pale17	52 ng/µl * 263 bp afl	57 ng/µl 192 bp afl	0,5 ng/µl 63 bp afl	7,9 ng/µl * 227 bp afl	5,5 ng/µl * 156 bp afl	44,8 ng/µl * 140 bp afl
Pale20	52ng/µl * 276 bp afl	147 ng/µl 242 bp afl	1 ng/µl 61 bp afl	7,3 ng/µl * 248 bp afl	5,1 ng/µl * 138 bp afl	42 ng/µl * 156 bp afl
Pale9	56 ng/µl * 200 bp afl	192 ng/µl 313 bp afl	2,1 ng/µl 92 bp afl	4,6 ng/µl * 168 bp afl	3,3 ng/µl * 114 bp afl	44,4 ng/µl * 111 bp afl
Smic37	29,9 ng/µl 291 bp afl	26,6 ng/µl 185 bp afl	5,6 ng/µl 47 bp afl	2,6 ng/µl - bp afl	0,5 ng/µl * 48 bp afl	127 ng/µl 65 bp afl
Smic41	22,4 ng/µl 210 bp afl	30,7 ng/µl 197 bp afl	8,4 ng/µl 57 bp afl	6,4 ng/µl - bp afl	0,7 ng/µl * 65 bp afl	168 ng/µl 50 bp afl
Sleb1	0,8 ng/µl - bp afl	0,3 ng/µl - bp afl	0,1 ng/µl - bp afl	too low - bp afl	too low * not measured	0,7 ng/µl * - bp afl
Imel1	2,2 ng/µl 103 bp afl	0,6 ng/µl 185 bp afl	0,1 ng/µl - bp afl	0,7 ng/µl 54 bp afl	too low * not measured	1,4 ng/µl * 43 bp afl
Aleu1	0,4 ng/µl 77 bp afl	0,4 ng/µl 75 bp afl	0,1 ng/µl - bp afl	too low - bp afl	too low * not measured	2 ng/µl * 48 bp afl
Smic39	0,8 ng/µl - bp afl	0,1 ng/µl 129 bp afl	0,2 ng/µl - bp afl	too low - bp afl	too low * not measured	3,4 ng/µl * not measured
Smic40	2,4 ng/µl * - bp afl	0,4 ng/µl 78 bp afl	too low - bp afl	too low - bp afl	too low * not measured	1,9 ng/µl * 49 bp afl

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Smic28	0,3 ng/µl - bp afl	0,1 ng/µl 68 bp afl	too low - bp afl	too low - bp afl	0,1 ng/µl * - bp afl	0,8 ng/µl * - bp afl
Smic29	0,7 ng/µl 60 bp afl	0,6 ng/µl 51 bp afl	too low - bp afl	too low - bp afl	0,1 ng/µl * - bp afl	1,8 ng/µl - bp afl
Smic31	34,6 ng/µl 179 bp afl	29,9 ng/µl 254 bp afl	7,9 ng/µl 53 bp afl	5,3 ng/µl 332 bp afl	3,2 ng/µl 64 bp afl	82,6 ng/µl 79 bp afl
Smic32	19,8 ng/µl 217 bp afl	39,7 ng/µl 98 bp afl	4,9 ng/µl 61 bp afl	5,0 ng/µl - bp afl	4,1 ng/µl 67 bp afl	157 ng/µl 64 bp afl
Smic33	30,7 ng/µl 193 bp afl	29,9 ng/µl 278 bp afl	5,1 ng/µl 56 bp afl	3,0 ng/µl 337 bp afl	2,8 ng/µl 62 bp afl	125 ng/µl 77 bp afl
Smic34	22,4 ng/µl 77 bp afl	8,7 ng/µl 81 bp afl	3,7 ng/µl 43 bp afl	1,5 ng/µl - bp afl	0,3 ng/µl * 84 bp afl	55 ng/µl 51 bp afl
Smic35	17,2 ng/µl 94 bp afl	16,3 ng/µl 79 bp afl	1,8 ng/µl 42 bp afl	1,9 ng/µl - bp afl	0,3 ng/µl * 2470? bp afl	51 ng/µl 44 bp afl
Smic38	17,9 ng/µl 187 bp afl	17,9 ng/µl 218 bp afl	2,8 ng/µl 40 bp afl	1,7 ng/µl - bp afl	no extraction	60 ng/µl 57 bp afl
Smic43	21,2 ng/µl 235 bp afl	0,2 ng/µl 116 bp afl	3,0 ng/µl 44 bp afl	2,7 ng/µl 281 bp afl	no extraction	214 ng/µl 92 bp afl
Smic45	19,6 ng/µl 246 bp afl	0,2 ng/µl 64 bp afl	5,1 ng/µl 40 bp afl	0,7 ng/µl - bp afl	no extraction	54,4 ng/µl 47 bp afl
Smic48	38 ng/µl 81 bp afl	0,2 ng/µl - bp afl	3,7 ng/µl 40 bp afl	3,9 ng/µl 256 bp afl	no extraction	126 ng/µl 58 bp afl
Smic36	18,6 ng/µl 90 bp afl	3,6 ng/µl 78 bp afl	3,6 ng/µl 45 bp afl	1,6 ng/µl 234 bp afl	no extraction	43,5 ng/µl 48 bp afl
Pale15	22,1 ng/µl * 230 bp afl	56 ng/µl 221 bp afl	2,1 ng/µl 61 bp afl	3,8 ng/µl * 187 bp afl	no extraction	37,6 ng/µl * 92 bp afl
Pale24	48,2 ng/µl 226 bp afl	120 ng/µl 271 bp afl	1,9 ng/µl 86 bp afl	5,3 ng/µl * 236 bp afl	no extraction	204 ng/µl * 122 bp afl
Pale64	20,2 ng/µl * 154 bp afl	10 ng/µl - bp afl	0,1 ng/µl - bp afl	1,5 ng/µl * 131 bp afl	no extraction	15 ng/µl * 83 bp afl
# 38	0,2 ng/µl - bp afl	11 ng/µl 288 bp afl	0,1 ng/µl - bp afl	too low * not measured	no extraction	4,3 ng/µl 63 bp afl
# 39	0,4 ng/µl - bp afl	6,9 ng/µl - bp afl	0,1 ng/µl - bp afl	too low * not measured	no extraction	2,6 ng/µl - bp afl
# 40	0,6 ng/µl 73 bp afl	15,5 ng/µl 82 bp afl	0,1 ng/µl - bp afl	too low * not measured	no extraction	4,7 ng/µl 495 bp afl
Cacu1	16,2 ng/µl 99 bp afl	14,3 ng/µl 93 bp afl	3,2 ng/µl 47 bp afl	1,0 ng/µl 89 bp afl	no extraction	69 ng/µl - bp afl
Lvir1	0,7 ng/µl - bp afl	0,4 ng/µl 113 bp afl	0,2 ng/µl - bp afl	0,1 ng/µl - bp afl	no extraction	6,7 ng/µl not measured
Poro1	17,7 ng/µl 172 bp afl	35,3 ng/µl 177 bp afl	5,7 ng/µl 66 bp afl	4,1 ng/µl 145 bp afl	no extraction	87 ng/µl 93 bp afl
Tmet1	3,4 ng/µl - bp afl	5,4 ng/µl 86 bp afl	1,2 ng/µl - bp afl	0,9 ng/µl 473 bp afl	no extraction	29 ng/µl * 85 bp afl
Tpol1	3,4 ng/µl 117 bp afl	3,2 ng/µl 117 bp afl	0,1 ng/µl - bp afl	0,2 ng/µl 87 bp afl	no extraction	1,3 ng/µl * 46 bp afl